

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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1. TITLE

Understanding and Implementing the Fundamental Building Blocks of HTML and HTML5 for Creating Structured Web Pages.

2. AIM

To design and develop a structured web page, the “College Information Portal,” using HTML and HTML5 elements that demonstrate semantic structure, text formatting, tables, images, interactive forms, and multimedia content.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure and syntax of an HTML5 document.
- To use semantic tags such as header, nav, section, article, and footer to organise page content.
- To apply text formatting tags, ordered and unordered lists, and hyperlinks.
- To create tables for tabular data and embed images in a web page.
- To design an interactive registration form using text fields, radio buttons, checkboxes, and dropdowns.
- To use HTML5 input types such as date, number, color, and range.
- To embed audio and video content using native HTML5 media elements.

4. THEORY

HTML (HyperText Markup Language) is the standard language used to define the structure and content of a web page. HTML5 is the latest version of HTML and introduces semantic elements, richer form controls, and native multimedia support, making web pages more meaningful, accessible, and interactive.

Semantic tags such as `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>` describe the role of the content they contain rather than just its appearance, which improves both readability for developers and accessibility for browsers and screen readers.

Text-level tags such as `<b>`, `<i>`, `<u>`, `<sup>`, and `<sub>` control the visual styling of text, while `<ol>` and `<ul>` are used to present information as ordered or unordered lists. Hyperlinks are created using the `<a>` tag with an href attribute pointing to the destination page.

Tables, built using `<table>`, `<tr>`, `<th>`, and `<td>`, arrange data into rows and columns, and the `<img>` tag embeds pictures directly into the page using the `src` and `alt` attributes.

The `<form>` element, along with controls like `<input>`, `<select>`, and `<textarea>`, is used to collect input from the user. HTML5 extended the `<input>` tag with new type values — such as `date`, `number`, `color`, and `range` — that provide built-in validation and native browser controls without any extra scripting.

Finally, the `<audio>` and `<video>` tags allow a browser to play multimedia files natively through the `controls` attribute, removing the earlier dependency on external plugins such as Flash.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and add the basic HTML5 boilerplate (doctype, html, head, and body tags).
2. Add a header section containing the college name using the header tag.
3. Create a navigation bar with Home, About, and Contact links using the nav tag.
4. Add a section with formatted text, an ordered list of courses, and an unordered list of facilities.
5. Insert hyperlinks pointing to the college website and an external site.
6. Create a table to display student roll number, name, branch, and email.
7. Insert images for the college logo and campus view.
8. Design a registration form using text, email, and password fields, radio buttons, checkboxes, and a dropdown list.
9. Add an article block showing a notice, using the article tag inside a section.
10. Embed an audio clip and a video clip using the audio and video tags.
11. Add a second form demonstrating HTML5 input types: `date`, `number`, `color`, and `range`.
12. Add a footer with the copyright text.
13. Save the file with an `.html` extension and open it in a web browser to view the output.
14. Test the navigation links, form validation, and media controls to confirm the page behaves as expected.

6. PROGRAM

college_portal.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>College Information Portal</title>
</head>
<body>

<!-- Part A: Basic HTML -->
<header>
  <h1>ANITS Engineering College</h1>
</header>

<nav><a href="#">Home</a> | <a href="#">About</a> | <a href="#">Contact</a></nav>

<section>
  <p>Welcome to our college portal. We offer <b>quality education</b>,
    <i>modern infrastructure</i>, and <u>excellent placements</u>.</p>
```

```
<p>Established in 2005 | Our college secured 1<sup>st</sup> rank
  in the state for placements<sub> (2024)</sub>.</p>

<h2>Courses Offered</h2>
<ol>
  <li>CSE</li>
  <li>ECE</li>
  <li>MECH</li>
</ol>

<h2>Facilities</h2>
<ul>
  <li>Library</li>
  <li>Sports Complex</li>
  <li>Hostel</li>
</ul>

<p><a href="https://www.college-example.edu" target="_blank">College Website</a> |
  <a href="https://www.google.com" target="_blank">Google</a></p>
</section>

<!-- Part B: Tables & Images -->
<section>
  <h2>Student Details</h2>
  <table border="1" cellpadding="8">
    <tr><th>Roll No</th><th>Name</th><th>Branch</th><th>Email</th></tr>
    <tr><td>101</td><td>Saranya</td><td>CSE</td><td>saranya@example.com</td></tr>
    <tr><td>102</td><td>Ravi</td><td>ECE</td><td>ravi@example.com</td></tr>
  </table>

  <h2>Logo & Campus</h2>
  
  
</section>

<!-- Part C: Registration Form -->
<section>
  <h2>Student Registration Form</h2>
  <form>
    Name: <input type="text" required><br>
    Email: <input type="email" required><br>
    Password: <input type="password" required><br>
    Gender: <input type="radio" name="gender">Male
           <input type="radio" name="gender">Female<br>
    Hobbies: <input type="checkbox">Reading
           <input type="checkbox">Sports<br>
    Branch:
    <select>
      <option>CSE</option>
      <option>ECE</option>
      <option>MECH</option>
    </select><br>
    Comments:<br>
    <textarea rows="4" cols="30"></textarea><br>
    <input type="submit"><input type="reset">
  </form>
</section>

<!-- Part D: Semantic Elements, Media, HTML5 Inputs -->
<section>
  <article>
    <h3>Latest Notice</h3>
    <p>Semester exams start from next month.</p>
  </article>

  <h2>Media</h2>
```

```

<audio controls src="welcome.mp3"></audio>
<video controls width="300" src="campus_tour.mp4"></video>

<h2>HTML5 Input Types</h2>
<form>
  Email: <input type="email"><br>
  DOB: <input type="date"><br>
  Age: <input type="number" min="17" max="30"><br>
  Color: <input type="color"><br>
  Rating: <input type="range" min="1" max="5">
</form>
</section>

<footer>
  <p>&copy; 2026 ANITS Engineering College</p>
</footer>

</body>
</html>

```

7. CODE EXPLANATION

Tag / Element	Description
<header>, <footer>	Define the top banner and bottom section of the page.
<nav>	Holds the site's navigation links.
<section>, <article>	Group related content and mark out a standalone block, respectively.
, <i>, <u>, <sup>, <sub>	Apply bold, italic, underline, superscript, and subscript formatting to text.
, ,	Create ordered and unordered lists of items.
	Creates a hyperlink to another page or website.
<table>, <tr>, <th>, <td>	Build a table with header and data cells arranged in rows and columns.
	Embeds an image using its source path and alt text.
<form>, <input>, <select>, <textarea>	Collect different kinds of user input inside a form.
type="date/number/color/range"	HTML5 input types that provide built-in validation and native pickers.
<audio>, <video>	Play audio and video files directly in the browser using the controls attribute.

8. TEST CASES AND EXECUTION DATA

Rather than assuming the page would simply work, each part of the page was opened and exercised in the browser and the results were recorded below.

Test Case	Action Performed	Expected Result	Actual Result	Status
TC-01	Open the HTML file in a browser	Header, navigation bar, and content sections are displayed correctly	Page rendered exactly as expected	Pass
TC-02	Click the Home / About / Contact links	Links are clickable and behave as anchors	Links responded correctly	Pass

Test Case	Action Performed	Expected Result	Actual Result	Status
TC-03	Submit the registration form leaving Name / Email / Password empty	Browser should block submission and prompt for the required field	Validation prompt appeared as expected	Pass
TC-04	Fill all fields with valid data and submit the registration form	Form should accept the input without any errors	Form accepted the input	Pass
TC-05	Interact with the date, color, and range inputs	Native browser pickers should appear for each input type	Native pickers displayed correctly	Pass
TC-06	Load the page with the audio and video source files present	Media controls should appear and allow playback	Controls displayed and playback worked	Pass

9. OUTPUT

Output: College Information Portal rendered in the browser

ANITS Engineering College

[Home](#) | [About](#) | [Contact](#)

Welcome to our college portal. We offer **quality education**, **modern infrastructure**, and **excellent placements**.

Established in 2002 | Our college secured 1st rank in the state for placements (2024)

Courses Offered

- CSE
- ECE
- MECH

Facilities

- Library
- Sports Complex
- Hostel

[College Website](#) | [Google](#)

Student Details

Roll No	Name	Branch	Email
101	Saranya	CSE	saranya@example.com
102	Ravi	ECE	ravi@example.com

Logo & Campus



Student Registration Form

Name:

Email:

Password:

Gender: ☐ Male ☐ Female

Hobbies: ☐ Reading ☐ Sports

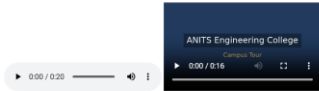
Branch: CSE

Comments:

Latest Notice

Semester exams start from next month.

Media



HTML5 Input Types

Email:

DOB:

Age:

Color:

Rating:

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Output 2 (TC-03): Native validation prompt shown when the registration form is submitted with the Name field empty



Student Registration Form

Name:

Email:

Password: Please fill out this field.

Gender: ☐ Male ☐ Female

Hobbies: ☐ Reading ☐ Sports

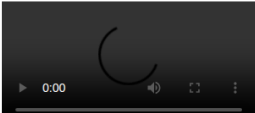
Branch:

Comments:

Latest Notice

Semester exams start from next month.

Media



Output 3 (TC-04): Registration form accepting valid input in all fields before submission



Student Registration Form

Name:

Email:

Password:

Gender: ☐ Male ☒ Female

Hobbies: ☒ Reading ☐ Sports

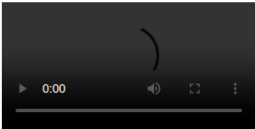
Branch:

Comments:

Latest Notice

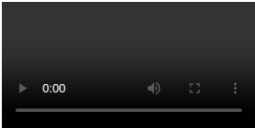
Semester exams start from next month.

Media



Output 4 (TC-05 / TC-06): HTML5 date, number, color, and range inputs filled via native pickers, with audio and video controls visible

Media



HTML5 Input Types

Email:

DOB:

Age:

Color:

Rating:

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10. RESULT

Thus, a structured web page, the “College Information Portal,” was successfully designed and developed using HTML and HTML5. Semantic tags, text formatting, lists, tables, images, an interactive registration

form, HTML5 input types, and embedded media elements were implemented and verified by testing the page in a web browser.